

1st Question:

Code:

```
import pandas as pd
```

```
# Read the Excel file
```

```
df = pd.read_excel("one.xlsx")
```

```
# Display the dataset
```

```
print("Departments Dataset:")
```

```
print(df)
```

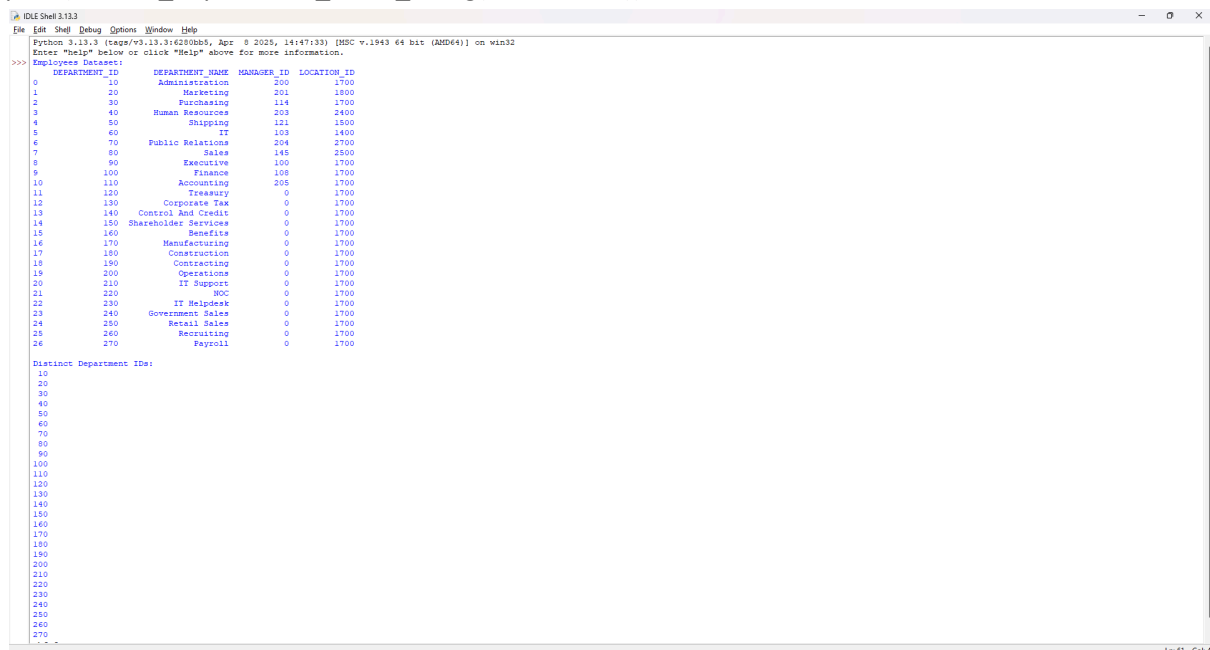
```
# Select distinct department IDs
```

```
distinct_department_ids = df["DEPARTMENT_ID"].drop_duplicates()
```

```
# Display the result
```

```
print("\nDistinct Department IDs:")
```

```
print(distinct_department_ids.to_string(index=False))
```



```
Python 3.13.3 (tags/v3.13.3:620bb5, Apr 8 2025, 14:47:13) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> Employees Dataset:
  DEPARTMENT_ID  DEPARTMENT_NAME  MANAGER_ID  LOCATION_ID
0             10      Administration        200           1700
1             20             Marketing        201           1800
2             30      Purchasing         114           1700
3             40      Human Resources        203           2400
4             50             Shipping        121           1800
5             60              IT          103           1400
6             70  Public Relations        204           2700
7             80             Sales         145           2500
8             90      Executive         100           1700
9            100             Finance         108           1700
10           110      Accounting         205           1700
11           120             Treasury          0           1700
12           130      Corporate Tax          0           1700
13           140  Control And Credit          0           1700
14           150  Shareholder Services          0           1700
15           160             Benefits          0           1700
16           170      Manufacturing          0           1700
17           180      Construction          0           1700
18           190      Contracting          0           1700
19           200      Operations          0           1700
20           210      IT Support          0           1700
21           220              MISC          0           1700
22           230      IT Helpdesk          0           1700
23           240  Government Sales          0           1700
24           250      Retail Sales          0           1700
25           260      Recruiting          0           1700
26           270      Payroll          0           1700

Distinct Department IDs:
10
20
30
40
50
60
70
80
90
100
110
120
130
140
150
160
170
180
190
200
210
220
230
240
250
260
270
```

2nd Question:

Code:

```
import pandas as pd
```

```
# Read Excel file
```

```
df = pd.read_excel("history.xlsx")
```

```
print("Employees History")
```

```
print(df)
```

```
# Employees who have done two or more jobs
```

```
result = df["EMPLOYEE_ID"].value_counts()
```

```
print("\nEmployees who have done two or more jobs:")
```

```
print(result[result >= 2].index.to_list())
```

```
===== RESTART: D:/query/exp 2.py =====
Employees History
  EMPLOYEE_ID  START DATE  END DATE  JOB_ID  DEPARTMENT_ID
0         102  2001-01-13  2006-07-24   IT_PROG             60
1         101  1997-09-21  2001-10-27  AC_ACCOUNT             110
2         101  2001-10-28  2005-03-15   AC_MGR             110
3         201  2004-02-17  2007-12-19   MK_REP             20
4        114  2006-03-24  2007-12-31  ST_CLERK             50
5        122  2007-01-01  2007-12-31  ST_CLERK             50
6        200  1995-09-17  2001-06-17  AD_ASST             90
7        176  2006-03-24  2006-12-31   SA_REP             80
8        176  2007-01-01  2007-12-31   SA_MAN             80
9        200  2002-07-01  2006-12-31  AC_ACCOUNT             90

Employees who have done two or more jobs:
[101, 200, 176]
> |
```

3rd question:

Code:

```
import pandas as pd
```

```
# Read Excel file
```

```
df = pd.read_excel("jobs.xlsx")
```

```
print("Jobs Dataset")
```

```
print(df)
```

```
# Sort by JOB_TITLE in descending order
```

```
result = df.sort_values(by="JOB_TITLE", ascending=False)
```

```
print("\nJobs in Descending Order of Job Title")
```

```
print(result)
```

Screenshot output:

```
===== RESTART: D:/query/exp 3.py =====
Jobs Dataset
   JOB_ID JOB_TITLE MIN_SALARY MAX_SALARY
0  AD_PRES  President    20080    40000
1    AD_VP  Administration Vice President    15000    30000
2  AD_ASST  Administration Assistant    3000     6000
3    FI_MGR   Finance Manager    8200    16000
4  FI_ACCOUNT  Accountant    4200     9000
5    AC_MGR  Accounting Manager    8200    16000
6  AC_ACCOUNT  Public Accountant    4200     9000
7    SA_MAN   Sales Manager    10000    20080
8    SA_REP  Sales Representative    6000    12008
9    PU_MAN  Purchasing Manager    8000    15000
10  PU_CLERK  Purchasing Clerk    2500     5500
11    ST_MAN   Stock Manager    5500     8500
12  ST_CLERK   Stock Clerk    2008     5000
13  SH_CLERK  Shipping Clerk    2500     5500
14  IT_PROG   Programmer    4000    10000
15    MK_MAN  Marketing Manager    9000    15000
16    MK_REP  Marketing Representative    4000     9000
17    HR_REP  Human Resources Representative    4000     9000
18    PR_REP  Public Relations Representative    4500    10500

Jobs in Descending Order of Job Title
   JOB_ID JOB_TITLE MIN_SALARY MAX_SALARY
11  ST_MAN   Stock Manager    5500     8500
12  ST_CLERK   Stock Clerk    2008     5000
13  SH_CLERK  Shipping Clerk    2500     5500
8    SA_REP  Sales Representative    6000    12008
7    SA_MAN   Sales Manager    10000    20080
9    PU_MAN  Purchasing Manager    8000    15000
10  PU_CLERK  Purchasing Clerk    2500     5500
18    PR_REP  Public Relations Representative    4500    10500
6  AC_ACCOUNT  Public Accountant    4200     9000
14  IT_PROG   Programmer    4000    10000
0  AD_PRES  President    20080    40000
16    MK_REP  Marketing Representative    4000     9000
15    MK_MAN  Marketing Manager    9000    15000
17    HR_REP  Human Resources Representative    4000     9000
3    FI_MGR   Finance Manager    8200    16000
1    AD_VP  Administration Vice President    15000    30000
2  AD_ASST  Administration Assistant    3000     6000
5    AC_MGR  Accounting Manager    8200    16000
4  FI_ACCOUNT  Accountant    4200     9000
```